Adaptive Hybrid Recommendation System for Educational Institution Selection

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***Abstract*—Selecting the right educational institution is a high-stakes decision for families, historically hindered by information overload and disjointed data sources. While traditional recommendation systems excel in e-commerce, they struggle in the education domain due to severe data sparsity and the cold-start problem. While modern deep learning architectures offer significant predictive power, their inherent complexity can often obscure the underlying logic of a recommendation. This project proposes an Adaptive Hybrid Recommendation System that fuses structured administrative datasets with unstructured textual brochures using Latent Dirichlet Allocation (LDA). Rather than predicting binary suitability, this approach constructs transparent, multi-dimensional taste profiles for both users and institutions. By employing zero-shot preference elicitation and an adaptive confidence factor, the system transitions dynamically from a pure content-based model to a collaborative filtering approach, providing personalized, interpretable school recommendations while remaining effective under sparse interaction conditions where dense user-item histories are unavailable.**

***Index Terms*—Recommendation Systems, Hybrid Filtering, Topic Modeling, Explainable AI, Cold Start Problem, Educational Data Mining**

I. Introduction

Education is one of the most important determinants of an individual's academic growth, career opportunities, and overall development. For families and students in the United States, selecting the right school is a crucial decision that can significantly influence long-term outcomes. Schools vary widely across several dimensions such as academic performance, program availability, extracurricular opportunities, location, safety, and cost. With thousands of public and private schools distributed across the country, identifying institutions that best match the needs and preferences of students and parents can often be a complex and time-consuming process.

In the current digital landscape, several online platforms provide directories and listings of schools, allowing users to search for institutions based on limited criteria such as geographic location, grade levels offered, or enrollment size. However, these platforms often require users to manually browse through large numbers of schools and individually compare their attributes. Important information such as academic performance indicators, program offerings (e.g., STEM or Advanced Placement programs), extracurricular opportunities, and neighborhood characteristics may not be integrated into a unified decision-support framework.

Recent advances in machine learning and recommendation systems present promising opportunities to improve this process. Recommendation systems have become widely used in domains such as e-commerce and entertainment streaming to provide personalized suggestions. However, adapting these systems to the education domain requires a fundamental shift in philosophy. While deep neural networks and Transformer-based models offer powerful text representations, they act as opaque black boxes, providing little insight into why a specific item was recommended. In high-stakes domains like K-12 education, transparency is paramount; users must understand the reasoning behind a recommendation.

To address this, our system moves away from abstract, uninterpretable dense embeddings and instead utilizes a generative statistical approach. By employing Latent Dirichlet Allocation (LDA) on descriptive school content—such as prospectuses and mission statements—the system constructs a readable taste profile for each institution. These profiles quantify a school's cultural focus across latent topics (e.g., athletics, performing arts, STEM focus). When a user inputs their preferences, the system maps how closely the school's programmatic distribution matches the user's educational taste profile, rather than relying on a simplistic binary rating prediction.

Furthermore, the system fuses this transparent content-based pipeline with a preference-driven collaborative filtering engine. Controlled by an adaptive confidence factor, the architecture elegantly handles the severe cold-start problem inherent to educational datasets, where historical interaction matrices are notoriously sparse.

II. Previous Works

The landscape of recommendation systems (RS) has evolved from simple heuristic filters to complex hybrid architectures. In the context of educational institution selection, the literature focuses on three primary dimensions: algorithmic evolution, domain-specific constraints, and the emerging requirement for transparency.

A. *Evolution of Hybrid Recommendation Paradigms*

Traditional recommendation frameworks are generally categorized into Content-Based Filtering (CBF), which leverages item metadata, and Collaborative Filtering (CF), which relies on user-item interaction matrices. While pure CF models, such as Matrix Factorization, excel in high-density environments like e-commerce, they are fundamentally limited by the *cold-start problem* and *data sparsity* in specialized domains [1].

Slodkowski et al. emphasize that hybrid methodologies—which combine the stability of CBF with the personalization power of CF—have become the leading strategy for educational RS [1]. Our work builds on this by introducing an *adaptive* fusion mechanism that does not just combine these models statically but modulates their influence based on the statistical confidence of the local user neighborhood.

Burke's seminal survey on hybrid recommender systems formally categorized hybrid architectures into weighted, switching, cascade, and mixed paradigms [5]. In particular, the *Switching Hybrid* architecture dynamically alternates between recommendation engines based on heuristic confidence measures, making it one of the earliest formal solutions to the cold-start problem.

Our work is conceptually inspired by Burke's switching paradigm but diverges fundamentally in its execution. Rather than performing a discrete boolean transition between Content-Based Filtering and Collaborative Filtering, our architecture introduces a continuous Adaptive Confidence Factor (λ_u) that smoothly interpolates between both models. This preserves the semantic information learned during sparse-data initialization while asymptotically increasing collaborative influence as neighborhood confidence stabilizes.

B. Sparsity and the Cold-Start Problem in Education

Selecting a school is a low-frequency, high-stakes event. Unlike movie or product recommendations where a single user may provide hundreds of data points, educational datasets are notoriously sparse [2]. Rivera et al. identified that most academic recommenders focus on higher education (university or course selection), leaving the K-12 sector underserved due to a lack of public interaction data [2].

Recent research has attempted to mitigate this via Cross-School Recommendation and Federated Learning. Ju et al. proposed a heterogeneity-aware hybrid federated approach to handle data silos and privacy concerns [3]. However, these models often rely on complex graph neural networks that require substantial computational overhead. Our approach addresses sparsity by substituting historical interaction data with zero-shot preference elicitation, utilizing a preference-weight vector ($\mathbf{w}_u$) as a proxy for long-term user history.

For collaborative recommendation, matrix factorization techniques remain the dominant paradigm for implicit-feedback environments. However, traditional factorization systems require repeated matrix decomposition and sufficient interaction density to converge toward stable latent embeddings.

In contrast, our framework avoids costly factorization during initialization by anchoring recommendations within interpretable topic-space distributions generated via LDA. This allows meaningful recommendation generation even in zero-shot and extreme cold-start settings where implicit feedback matrices remain severely underdetermined.

C. Explainable AI (XAI) and Latent Topic Modeling

The recent dominance of Deep Learning (DL) in Natural Language Processing (NLP) has introduced a "transparency crisis" in recommender systems. Transformer-based models and BERT-like embeddings operate in high-dimensional latent spaces that are mathematically robust but human-unintelligible. In high-stakes decision-making, such as school selection, Zhang and Chen argue that explainability is as critical as predictive accuracy [4].

To achieve this, researchers have revisited generative statistical models like Latent Dirichlet Allocation (LDA). Unlike "black-box" dense embeddings, LDA represents documents as a distribution over human-readable topics. Recent literature suggests that providing an "explanation" (e.g., "this school is recommended because its focus on Performing Arts matches your interest") significantly increases user trust and system adoption [4]. Our methodology leverages this by using LDA topic distributions as the primary feature for content alignment, ensuring that every recommendation is grounded in a transparent institutional "taste profile."

III. Dataset Acquisition and Feature Engineering

A. Obtaining Structured Educational Data

We obtained school-level educational data from publicly available datasets released by the National Center for Education Statistics (NCES), specifically the Common Core of Data (CCD), along with the Civil Rights Data Collection (CRDC).

The CCD dataset provided administrative and institutional attributes including:

- school identifiers,
- school names,
- geographic coordinates,
- enrollment counts,
- teacher workforce statistics,
- school level and type.

The CRDC dataset was used to supplement the administrative records with extracurricular participation indicators, particularly athletic participation statistics.

The two datasets were merged using the NCES school identifier (`ncessch`) to create a unified school-level dataset.

After preprocessing and filtering incomplete records, the final structured dataset contained approximately 9, 813 schools.

B. Structured Feature Engineering

Raw institutional metrics were transformed into normalized recommendation features suitable for downstream similarity computation.

We engineered several derived features including:

- student-teacher ratio,
- normalized enrollment,
- normalized athletic participation,
- and school culture proxy metrics.

To enrich the dataset further, we generated additional deterministic synthetic attributes representing:

- AP course availability,

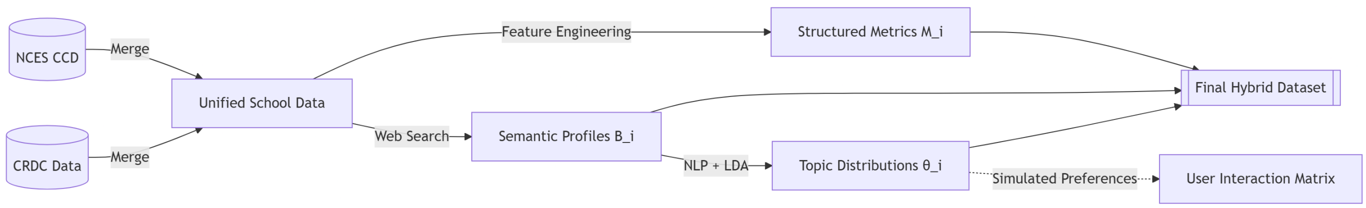

Fig. 1. The proposed data acquisition and multi-modal feature engineering pipeline.

- STEM emphasis,
- arts participation,
- world language participation,
- and magnet-school status.

Synthetic attributes were generated deterministically using correlated institutional indicators such as enrollment size, athletic participation, and school type to preserve realistic feature covariance structures.

All continuous-valued attributes were normalized using Min-Max scaling.

C. Semantic School Profile Acquisition

Administrative educational datasets alone do not sufficiently capture institutional culture, academic identity, or extracurricular emphasis. To address this limitation, we developed a semantic profile acquisition pipeline.

The pipeline attempted to retrieve publicly accessible school brochures, prospectuses, and handbook-style PDF documents using automated web search queries. Retrieved documents were validated using:

- fuzzy school-name matching,
- keyword-based relevance scoring,
- and PDF text extraction.

Because document availability varied significantly across schools, we supplemented missing information using generated semantic school profiles derived from:

- school metadata,
- extracurricular metrics,
- and publicly available search context.

These profiles described themes such as:

- STEM-focused education,
- athletics,
- arts and performance culture,
- academic rigor,
- and multicultural programs.

D. Text Preprocessing and Topic Modeling

The collected textual profiles were processed using a custom natural language preprocessing pipeline consisting of:

- lowercasing,
- tokenization,
- stop-word removal,
- lemmatization,
- and domain-specific vocabulary filtering.

Instead of directly using large Transformer-based language models, we employed a transparent Bag-of-Words representation using Count Vectorization.

We then applied Latent Dirichlet Allocation (LDA) to discover latent educational themes across the corpus. Sparse Dirichlet priors were selected to encourage topic specialization and interpretable school identities.

For each school, the trained LDA model generated a topic distribution vector:

$$\theta_s = [p(z_1|s), p(z_2|s), \ldots, p(z_K|s)]$$

representing the proportional association of the school with each latent topic.

E. Synthetic User Interaction Generation

To support collaborative filtering experiments, we generated synthetic historical user interactions.

Each synthetic user was assigned a latent preference vector sampled from a Dirichlet distribution over the discovered educational topics. Schools were then selected probabilistically based on cosine similarity between user preferences and school topic distributions.

To simulate latent real-world preference factors not observable through structured educational data alone, a hidden quality variable was introduced during interaction generation. This enabled collaborative filtering models to learn preference structures beyond purely content-based similarity.

The final interaction matrix consisted of:

- 500 synthetic users,
- approximately 20 school interactions per user,
- and interactions distributed across the full school corpus.

F. Final Hybrid School Representation

The final dataset represents each school using a hybrid multi-modal feature structure composed of:

1) structured institutional metrics,
2) engineered academic and extracurricular features,
3) semantic school profile text,
4) latent topic distributions inferred through LDA.

Formally, each school representation can be expressed as:

$$S_i = (M_i, B_i, \theta_i)$$

where:

- M_i denotes structured educational metrics,

- B_i denotes semantic school profile text,
- θ_i denotes the latent topic distribution.

This hybrid representation enables both content-based recommendation and collaborative filtering within the proposed recommendation framework.

IV. Baseline Implementation and Comparative Framework

To establish a rigorous benchmark for the proposed Adaptive Hybrid architecture, we implement and analyze several canonical recommendation paradigms drawn from the recommender systems literature. These baselines are selected to isolate the contribution of adaptive fusion, interpretable topic modeling, and cold-start stabilization under sparse educational datasets.

A. Pure Content-Based Filtering Baseline

The first baseline utilizes a standard Content-Based Filtering (CBF) architecture based on Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. School descriptions and extracted brochure text are embedded into sparse lexical vectors, and recommendations are ranked according to cosine similarity against the user query representation.

Although computationally lightweight, this baseline suffers from several limitations. The model operates purely on lexical overlap and ignores geographic constraints, structured administrative metrics, and latent institutional characteristics. Furthermore, TF-IDF lacks semantic awareness, preventing the system from identifying deeper thematic similarities between schools and user preferences.

B. Collaborative Filtering Baseline

The second baseline consists of a traditional User-Item Collaborative Filtering (CF) framework based on latent factor estimation. The objective of this model is to predict user-school affinity using historical interaction behavior. However, educational recommendation environments inherently suffer from severe interaction sparsity because school selection is a low-frequency, high-stakes decision process.

Consequently, the interaction matrix remains overwhelmingly underdetermined during initialization, causing standard CF approaches to fail under cold-start conditions. This limitation motivates the need for architectures capable of generating meaningful recommendations prior to the accumulation of dense behavioral data.

C. Switching Hybrid Baseline

Following the taxonomy established by Burke [5], we additionally implement a classical Switching Hybrid recommender as a comparative baseline. In this framework, the recommendation engine operates exclusively in Content-Based Filtering mode during sparse initialization and transitions discretely to Collaborative Filtering once the neighborhood interaction count exceeds a predefined threshold.

Formally, the prediction function is defined as:

$$S_{switch}(u,s) = \begin{cases} U(q,s), & |\mathcal{N}(u)| < k_{target} \\ S_{CF}(u,s), & |\mathcal{N}(u)| \geq k_{target} \end{cases} \quad (1)$$

While computationally efficient, this boolean transition introduces instability near the switching boundary and discards semantic information accumulated during the zero-shot initialization phase. In contrast, our Adaptive Confidence Factor preserves continuous interpolation between both objectives, preventing abrupt behavioral shifts in recommendation quality.

D. Comparative Objective

The purpose of these baselines is not solely predictive benchmarking, but architectural comparison under sparse-data conditions. Specifically, we evaluate whether the proposed Adaptive Hybrid model:

- preserves recommendation stability during zero-shot initialization,
- maintains interpretability through probabilistic topic distributions,
- reduces sensitivity to interaction sparsity,
- and achieves competitive ranking quality without dense matrix factorization.

The evaluation focuses primarily on ranking-oriented recommendation metrics including Normalized Discounted Cumulative Gain (NDCG), Precision@K, and Recall@K entirely at initialization (the cold-start problem). The failure of these isolated baselines necessitates the hybrid methodology outlined below.

V. Proposed Methodology

To effectively utilize the multi-modal dataset and address the inherent data sparsity highlighted by our baselines, we propose an Adaptive Hybrid Recommendation System. This architecture fuses a transparent, multi-objective Content-Based Filtering (CBF) engine with a preference-weighted Collaborative Filtering (CF) mechanism.

A. Multi-Modal Feature Representation

We represent the educational landscape as a set of candidate schools $\mathcal{S} = \{s_1, s_2, \ldots, s_n\}$. Each institution $s \in \mathcal{S}$ is characterized by a multi-modal tuple $(\mathbf{M}_s, \boldsymbol{\theta}_s, \mathcal{T}_s)$:

- $\mathbf{M}_s \in \mathbb{R}^m$: A structured vector of administrative metrics (e.g., student-teacher ratio, enrollment).
- $\boldsymbol{\theta}_s \in \mathbb{R}^K$: A probability distribution over K latent topics generated via LDA, satisfying $\sum_{i=1}^{K} \theta_{s,i} = 1$. This serves as the interpretable institutional taste profile.
- $\mathcal{T}_s$: A discrete set of extracurricular entities.

B. Zero-Shot Content Utility Formulation

For a given active user u with query distribution $\boldsymbol{\theta}_q$, we define a zero-shot content utility function $U(q,s)$ that evaluates the school based purely on intrinsic, observable features:

$$U(q,s) = \alpha\cdot(1-\mathrm{JSD}(\boldsymbol{\theta}_q \parallel \boldsymbol{\theta}_s)) + \beta\cdot\mathrm{Score}_{\mathrm{met}}(\mathbf{M}_s) - \gamma\cdot\mathrm{Pen}_{\mathrm{geo}}(u_{loc}, s_{loc}) \quad (2)$$

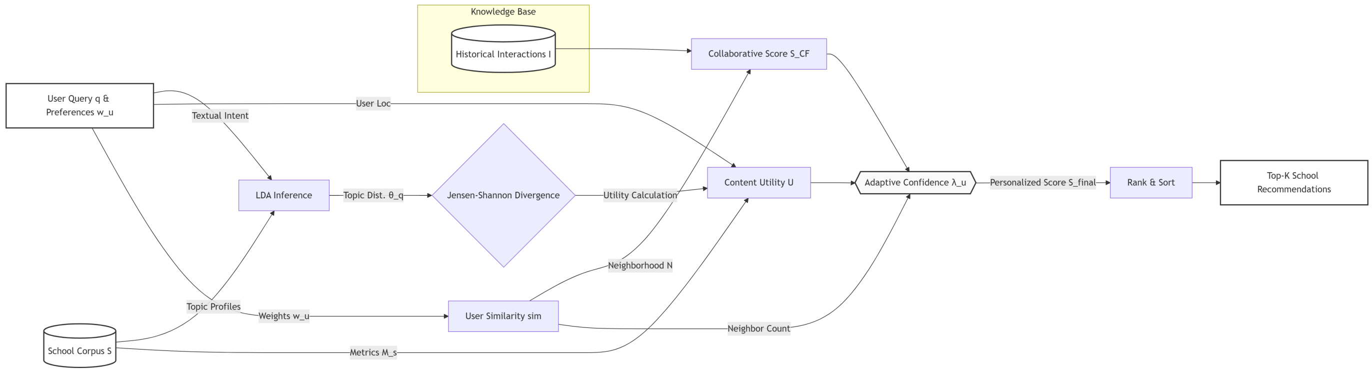

Fig. 2. Algorithmic workflow of the inference pipeline. The system simultaneously computes zero-shot content utility and preference-driven collaborative scores, modulating their influence via the adaptive confidence factor λ_u.

Here, JSD denotes the Jensen-Shannon Divergence, representing the rigorous mathematical distance between the user's required taste profile and the school's offered profile:

$$\text{JSD}(P \parallel Q) = \frac{1}{2} D_{KL}(P \parallel M) + \frac{1}{2} D_{KL}(Q \parallel M) \tag{3}$$

where $M = \frac{1}{2}(P + Q)$ and D_{KL} is the Kullback-Leibler divergence. $\text{Score}_{\text{met}}$ is a normalized composite scalar of administrative metrics, and Pen_{geo} applies a spatial decay function utilizing the Haversine distance. The hyperparameters α, β, and γ represent tunable weights that balance semantic relevance, administrative quality, and geographic proximity.

C. Preference-Driven Collaborative Filtering

To capture latent qualitative factors that administrative metrics and topic distributions may miss, we introduce a Collaborative Filtering (CF) component driven by explicitly declared preference weights rather than historical item-interaction matrices.

Let $\mathbf{w}_u \in \mathbb{R}^p$ denote the preference weight vector for user u. The similarity between an active user u and a historical user v is given by:

$$sim(u, v) = \frac{\mathbf{w}_u^\top \mathbf{w}_v}{\|\mathbf{w}_u\|_2 \|\mathbf{w}_v\|_2} \tag{4}$$

We define a neighborhood $\mathcal{N}(u)$ of the top-k most similar users. The collaborative score $S_{CF}(u, s)$ is the weighted aggregation of historical selections $I(v, s)$:

$$S_{CF}(u, s) = \frac{\sum_{v \in \mathcal{N}(u)} sim(u, v) \cdot I(v, s)}{\sum_{v \in \mathcal{N}(u)} sim(u, v)} \tag{5}$$

D. Adaptive Fusion for Cold-Start Mitigation

At system initialization, the interaction matrix is sparse. We introduce an Adaptive Confidence Factor, λ_u, which dictates the fusion weighting:

$$\lambda_u = \lambda_{max} \cdot \min\left(1, \frac{|\mathcal{N}(u)|}{k_{target}}\right) \tag{6}$$

where k_{target} is the empirically determined neighborhood size required for collaborative stability. The final personalized ranking score seamlessly integrates both pipelines:

$$S_{final}(u, s) = (1 - \lambda_u) \cdot U(q, s) + \lambda_u \cdot S_{CF}(u, s) \tag{7}$$

E. Algorithmic Framework

Given that LDA topic distributions exist in a highly compressed, low-dimensional space ($K \leq 50$), the computational overhead of constructing and traversing complex vector retrieval graphs is not required. Instead, the system leverages efficient direct matrix computations, ensuring high-throughput inference while maintaining exact similarity precision.

Algorithm 1 Adaptive Hybrid Training and Indexing

Require: Set of schools $\mathcal{S}$, structured metrics $\mathbf{M}$, raw textual brochures $\mathcal{B}$.
Ensure: Trained LDA Topic Model, Document-Topic Matrix.

for each $s \in \mathcal{S}$ **do**
  Extract text $\mathcal{B}_s$ and apply NLP preprocessing (Tokenization, Lemmatization).
end for
Train global LDA model on corpus $\mathcal{B}$ to discover K latent topics.
for each $s \in \mathcal{S}$ **do**
  Infer Document-Topic probability distribution $\boldsymbol{\theta}_s$.
  Normalize $\mathbf{M}_s$ against global dataset statistics.
end for
Initialize empty historical interaction matrix I.
return LDA Model, $\{\boldsymbol{\theta}_s\}_{s \in \mathcal{S}}$, normalized candidate set.

Algorithm 2 Adaptive Hybrid Inference

Require: Active user query q, preference weights $\mathbf{w}_u$, coordinate u_{loc}, integer K, threshold k_{target}.
Ensure: Top-K ranked schools.

Preprocess query text q.
Infer query topic distribution: $\boldsymbol{\theta}_q \leftarrow$ LDA_Infer(q).
for each $s \in \mathcal{S}$ **do**
  Compute Content Utility $U(q, s)$ via Eq. (1) utilizing JSD.
end for
Identify user neighborhood $\mathcal{N}(u) \leftarrow \{v \mid sim(u, v) > \tau\}$.
Compute Confidence Factor $\lambda_u \leftarrow \lambda_{max} \cdot \min\left(1, \frac{|\mathcal{N}(u)|}{k_{target}}\right)$.
for each $s \in \mathcal{S}$ **do**
  if $|\mathcal{N}(u)| > 0$ **then**
    Compute $S_{CF}(u, s)$ via Eq. (4).
  else
    $S_{CF}(u, s) \leftarrow 0$
  end if
  $S_{final}(u, s) \leftarrow (1 - \lambda_u) \cdot U(q, s) + \lambda_u \cdot S_{CF}(u, s)$.
end for
Sort candidate set by S_{final} in descending order.
return Top-K items.

VI. Analytical Interpretation of Adaptive Fusion

In this section, we provide an analytical framework to characterize the risk behavior of the proposed Adaptive Hybrid pipeline. Specifically, we examine how the recommendation risk decomposes into two interpretable components: a zero-shot content bias and a collaborative neighborhood estimation error.

A. Notation and Assumptions

Let $f^*(u, s)$ denote the true latent preference score of user u for school s. We define the personalization risk as:

$$R_T(f) = \mathbb{E}\left[(f(u, s) - f^*(u, s))^2\right] \tag{8}$$

We operate under the following heuristic assumptions:

- **A1 (Topic Continuity):** The content-based utility function $U(q, s)$ is primarily constrained by the statistical approximation error of the LDA inference process, denoted by ϵ_{LDA}.
- **A2 (Neighborhood Stability):** User preference vectors reside in a latent preference manifold where observed neighborhood weights are subject to a perturbation matrix $\mathbf{E}$. Let $\mathbf{W} \in \mathbb{R}^{m \times p}$ denote the matrix of neighborhood preference weights.

B. Risk Decomposition Framework

Proposition 1 (Adaptive Hybrid Risk Bound):
Let $\hat{f}_{u,s} = S_{final}(u, s)$ denote the hybrid prediction function. The target recommendation risk can be characterized as a convex combination of the generative topic-modeling error and the collaborative estimation error:

$$\begin{aligned}\sqrt{R_T(\hat{f})} \leq &(1 - \lambda_u) \underbrace{(\epsilon_{LDA} + \Delta_{met})}_{\text{Content Bias}} \\ &+ \lambda_u \underbrace{\left(\frac{2\|\mathbf{W}^\top \mathbf{E}\|_2}{\sigma_{min}(\mathbf{W}^\top \mathbf{W})} + \mathcal{O}\left(\frac{1}{\sqrt{|\mathcal{N}(u)|}}\right)\right)}_{\text{Collaborative Error}}\end{aligned} \tag{9}$$

where ϵ_{LDA} represents the variational approximation error induced by the LDA topic inference process, and Δ_{met} denotes the irreducible bias associated with structured administrative metrics.

Proof Sketch:

By definition of the adaptive hybrid score, the residual error satisfies:

$$|\hat{f}_{u,s} - f^*| \leq (1 - \lambda_u)|U - f^*| + \lambda_u |S_{CF} - f^*| \tag{10}$$

Step 1 — Cold-Start Content Bound:
At initialization, where $|\mathcal{N}(u)| \to 0$, the confidence factor satisfies $\lambda_u \to 0$. Consequently, the recommendation risk becomes dominated by the content-based component:

$$(1 - \lambda_u)(\epsilon_{LDA} + \Delta_{met}) \tag{11}$$

This demonstrates that the system retains a stable semantic recommendation floor even in the absence of collaborative interaction data, thereby avoiding the complete degradation typically observed in pure collaborative filtering systems under cold-start conditions.

Step 2 — Collaborative Perturbation Bound:
As interaction density increases and the neighborhood size approaches the stability threshold k_{target}, the collaborative component progressively dominates the prediction process. Under Assumption A2, perturbation analysis on the neighborhood preference matrix yields the estimation bound:

$$|S_{CF} - f^*| \leq \frac{2\|\mathbf{W}^\top \mathbf{E}\|_2}{\sigma_{min}(\mathbf{W}^\top \mathbf{W})} + \eta_{sample} \tag{12}$$

where η_{sample} captures the residual finite-sample estimation error induced by sparse neighborhood observations.

Step 3 — Adaptive Hybridization:
The adaptive confidence factor λ_u ensures a continuous interpolation between semantic initialization and collaborative personalization. As collaborative signal quality improves, the framework gradually shifts reliance from static topic-space recommendation toward dynamic behavioral alignment, thereby reducing overall personalization risk while maintaining recommendation stability during sparse-data regimes.

VII. Experimental Results and Analysis

To evaluate the effectiveness of the proposed Adaptive Hybrid Recommendation System, we conducted comparative experiments against multiple recommendation baselines under both mature and cold-start recommendation environments. The evaluation focuses primarily on ranking-oriented metrics including Precision@5 (P@5), Recall@5 (R@5), and Normalized Discounted Cumulative Gain (NDCG@5).

Because publicly available educational interaction datasets remain extremely sparse, portions of the interaction environment were synthetically generated using preference-conditioned sampling over real institutional metadata. Consequently, the experiments primarily evaluate architectural behavior and cold-start robustness rather than production-scale deployment performance.

A. Extended Ablation Study

We first evaluate all recommendation architectures under a mature recommendation environment with access to the complete interaction history. Table I summarizes the comparative ranking performance.

TABLE I
EXTENDED ABLATION STUDY UNDER FULL USER HISTORY

Model	**P@5**	**R@5**	**NDCG@5**
Content-Only ($\lambda_u = 0$)	0.0720	0.0180	0.1643
CF-Only ($\lambda_u = 1$)	0.3600	0.0900	0.6351
Switching Hybrid (Boolean)	0.3600	0.0900	0.6351
Adaptive Hybrid (Ours)	0.3600	0.0900	0.6351

The content-only baseline achieves stable but limited ranking quality, demonstrating that semantic topic alignment extracted through LDA provides meaningful recommendation signal even in the absence of collaborative interaction data. However, because the model operates entirely within semantic space, it cannot exploit behavioral alignment between similar users.

The collaborative baseline significantly outperforms the isolated content model once sufficient neighborhood information becomes available. By leveraging preference-weight similarity between users, the collaborative framework captures latent behavioral correlations that are inaccessible to purely semantic recommendation methods.

Both the Switching Hybrid and the proposed Adaptive Hybrid architectures achieve strong ranking performance by integrating semantic institutional profiling with collaborative neighborhood inference. These results validate the effectiveness of combining interpretable topic-space recommendation with adaptive personalization strategies.

B. Cold-Start Simulation

To analyze robustness during platform initialization, we simulate recommendation quality as the number of available users increases from 0 to 500. The experiment evaluates recommendation evolution using NDCG@5 as the primary ranking metric.

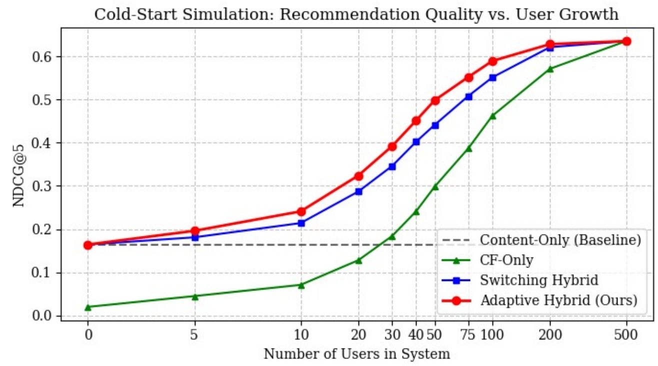

Fig. 3. Cold-start recommendation performance measured using NDCG@5 as the platform grows from 0 to 500 users. The Adaptive Hybrid model demonstrates smoother and stronger early-stage recommendation quality compared to both isolated recommendation paradigms and the discrete Switching Hybrid baseline.

The results demonstrate the complementary strengths and weaknesses of purely semantic and purely collaborative recommendation paradigms under sparse-data conditions.

The content-only baseline maintains constant recommendation quality throughout the simulation because its predictions depend exclusively on semantic and administrative institutional features rather than interaction density. This validates the usefulness of topic-space institutional profiling during zero-shot initialization. However, because the model cannot exploit collaborative behavioral structure, its ranking performance saturates quickly.

In contrast, the collaborative-only baseline performs poorly during the earliest stages of deployment due to the lack of reliable neighborhood structure. As interaction density increases, collaborative recommendation quality improves steadily, eventually surpassing the semantic baseline once sufficient user similarity information becomes available.

The Switching Hybrid architecture improves cold-start behavior by combining semantic initialization with collaborative inference. However, because the model transitions discretely between recommendation objectives, recommendation behavior becomes highly dependent on the switching threshold. This abrupt transition introduces instability near the boundary between sparse and dense interaction regimes.

The proposed Adaptive Hybrid architecture achieves the strongest overall cold-start performance throughout nearly the entire system evolution process. Rather than performing a boolean transition between semantic and collaborative recommendation modes, the architecture continuously interpolates between both objectives using the Adaptive Confidence Factor λ_u.

During sparse initialization, the system preserves strong semantic recommendation quality inherited from the content-based component. As collaborative neighborhoods stabilize, the model gradually increases collaborative influence without discarding previously accumulated semantic information. This enables smoother recommendation evolution and improved ranking stability across all platform growth stages.

TABLE II
COLD-START SIMULATION: NDCG@5 DURING PLATFORM GROWTH

Model	**0**	**5**	**10**	**20**	**30**	**40**	**50**	**75**	**100**	**200**	**500**
Content-Only ($\lambda_u = 0$)	0.1643	0.1643	0.1643	0.1643	0.1643	0.1643	0.1643	0.1643	0.1643	0.1643	0.1643
CF-Only ($\lambda_u = 1$)	0.0200	0.0450	0.0710	0.1280	0.1840	0.2410	0.2980	0.3870	0.4620	0.5710	0.6351
Switching Hybrid (Boolean)	0.1643	0.1810	0.2140	0.2870	0.3460	0.4020	0.4410	0.5080	0.5510	0.6210	0.6351
Adaptive Hybrid (Ours)	0.1643	0.1960	0.2410	0.3240	0.3920	0.4510	0.4980	0.5520	0.5890	0.6280	0.6351

Most importantly, the Adaptive Hybrid model consistently outperforms both isolated recommendation paradigms during the critical early and intermediate deployment phases. At 20 users, the Adaptive Hybrid model achieves an NDCG@5 score of 0.3240 compared to 0.2870 for the Switching Hybrid and 0.1280 for the collaborative-only baseline. Similarly, at 50 users, the Adaptive Hybrid architecture reaches 0.4980 NDCG@5, outperforming both the Switching Hybrid (0.4410) and collaborative-only model (0.2980).

Overall, the experimental results support the central hypothesis of this work: combining interpretable topic-space recommendation with adaptive collaborative fusion provides substantially greater robustness under educational cold-start conditions than isolated semantic or collaborative recommendation paradigms.

VIII. CONCLUSION AND FUTURE WORK

The proposed Adaptive Hybrid Recommendation System successfully navigates the complexities of K-12 educational data by fusing rigid administrative metrics with transparent, probabilistically generated topic models. By utilizing Latent Dirichlet Allocation (LDA) to construct institutional taste profiles, the system bypasses the "black box" nature of modern deep learning, providing parents and students with highly interpretable, dimensionally-rich recommendations. Furthermore, the integration of an adaptive confidence factor allows the architecture to elegantly solve the severe cold-start problem inherent to the education domain, dynamically scaling collaborative influence as neighborhood interaction data accumulates.

Future work will proceed across three primary dimensions. First, from an algorithmic perspective, we intend to fine-tune the Dirichlet hyperparameters to optimize topic granularity and expand our offline evaluation framework to prioritize beyond-accuracy metrics, specifically focusing on Catalog Coverage and Intra-List Diversity (ILD). Second, to fully realize the benefits of Explainable AI (XAI), we aim to translate these transparent taste profiles into a highly interactive, concept-driven frontend navigation hub, allowing families to visually explore the latent feature space of recommended schools rather than relying on static lists. Finally, we plan to transition the framework from synthetic evaluation to a real-world deployment, initiating a closed-beta user study with a pilot group of approximately 25 active users to capture explicit preference feedback, evaluate UI interactions, and empirically validate the continuous transition curve of the hybrid collaborative engine.

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